

2022/2023

01

Rocks in the Brain

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Location
US

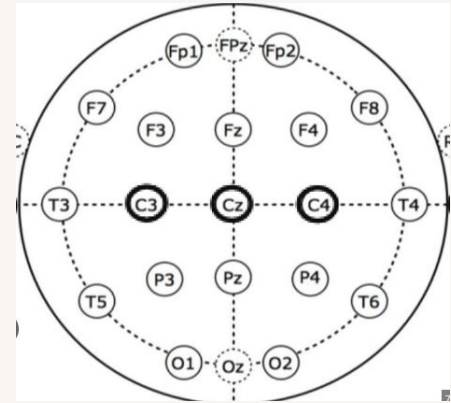
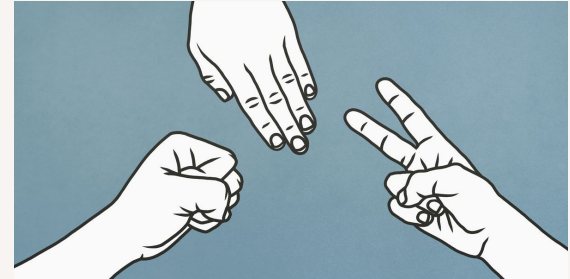
Project Time
2 weeks

My Role

While everyone in the group worked on all parts of the project, I was the person in charge of conducting the experiment trials and improving the algorithm.

Project Brief

This is the final project of COGS 189, taught by Dr. Alessandro D'Amico. In this project, we designed experiment trials and collected EEG data to build a classification model to predict hand movements for use in real time play of rock-paper scissor.



Overview

Motivation

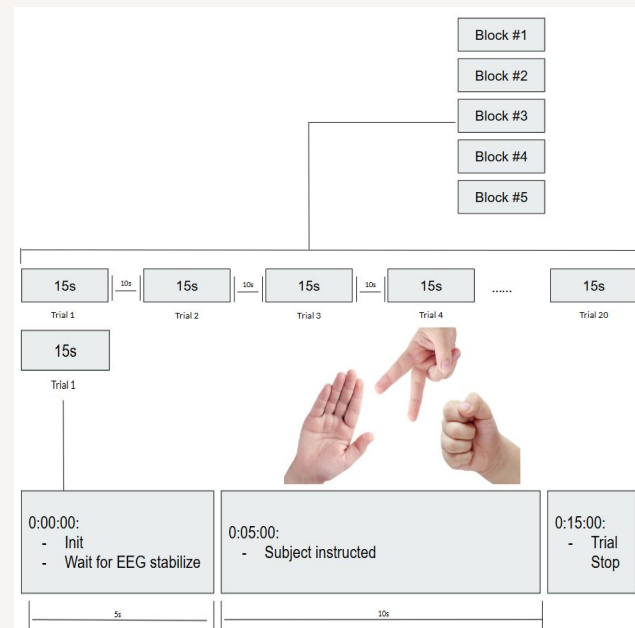
The goal of this project is to learn about how to work with brain-computer interfaces (BCIs) and to increase our understanding of motor control. This project can serve as an introduction to behavioral science, and pave way for BCI-controlled prosthetics, therapeutic technology, and other technologies that can aid or even enhance human cognition.

Method

Data collection

The subject plays rock-paper-scissor against a groupmate by imagining the hand gesture without making it

- 20 trials x 5 blocks
- 15 seconds/trial and 10 second break in between



Algorithm

Used a simple neural network

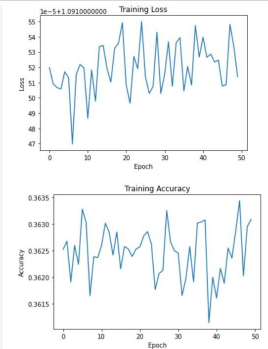
- Two hidden layers, three neurons each
- Sigmoid activation function
- 50 epochs
- Error metrics: Accuracy $(TP+TN)/(TP+FP+TN+FN)$
- Test accuracy: 0.3619

```
In [18]: from tensorflow.keras.models import Sequential  
        from tensorflow.keras.layers import Dense
```

```
In [19]: model = Sequential()  
        model.add(Dense(3, activation='sigmoid'))  
        model.add(Dense(3, activation='sigmoid'))
```

```
In [22]: model.compile(loss='categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
```

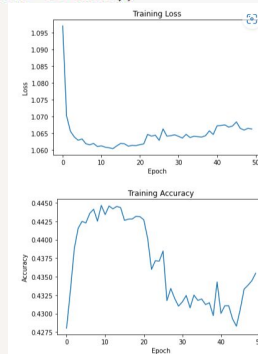
```
In [24]: history = model.fit(X_train_tensor, y_train_tensor, epochs=50, verbose=1)
```



ReLU activation function (popular non-linear function for large datasets)

- Takes much longer to train
- 50 epochs
- Test accuracy: 0.401
- Added batch normalization
- Added a dropout of .5
- Test accuracy: 0.440

```
model = Sequential()  
model.add(Dense(64, activation='relu', input_shape=(3,), kernel_regularizer=l2(0.01)))  
model.add(BatchNormalization())  
model.add(Dense(32, activation='relu', kernel_regularizer=l2(0.01)))  
model.add(Dropout(0.5))  
model.add(Dense(3, activation='softmax'))
```



Conclusion

- Highest test accuracy we could obtain was 44%
- Tried other methods (SVM using one vs one and one vs rest, random forest) but it took too long to train so they were not viable

Improvements to make

- Get more experience with collecting data through EEG
 - Better data with less artifacts and other interference
 - Adjustments to our data collecting method
- More data
 - Very limited time with it to collect our data due to EEG caps being shared between groups
- A more powerful machine to collect and process the data
 - Laptop was overheating throughout the trial
 - It was taking too long to train and run the models

Takeaways

- Experience working with BCIs
- Experience building neural network and other algorithms such as ReLU and random forest
- Designing experiment to measure data for behavioral science